Vivek Bhavsar

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Summary

Principal Data Scientist and Head of AI Research & Data Science with a PhD in Computational and Applied Mathematics and 6+ years of experience in LLMs, agentic systems, and model evaluation for high-stakes domains. Built and lead a first-of-its-kind agentic AI platform for prosecutors, now in production on Azure Government, that **improved case-preparation efficiency by more than 80%** for cases with 5,000+ files and higher complexity. Research focus: workflow-level evaluation of legal agents (AAAI-27 manuscript), claim-level hallucination metrics, calibrated document understanding with abstention, and human-in-the-loop studies with practicing attorneys. Interested in research on LLM reliability, evaluation, and safe deployment.

Technical Skills

LLMs & Agents: agentic workflows, tool use, Model Context Protocol (MCP), OpenAI Agents SDK, RAG (hybrid retrieval, cross-encoder reranking), text-to-SQL, LLM-as-judge, context engineering, Goverment Cloud Azure OpenAI
Research & Evaluation: claim-level hallucination metrics, calibration and selective prediction, counterfactual fault injection, preregistered human studies, paired bootstrap testing, document AI (OCR, layout models), NER/PII redaction
ML Frameworks: PyTorch, JAX, TensorFlow, Hugging Face Transformers, sentence-transformers, DeepSpeed ZeRO-3, ONNX Runtime, CUDA, scikit-learn, XGBoost, FAISS
Languages & Data: Python, SQL/T-SQL, C/C++, TypeScript, Bash, MATLAB
Cloud & MLOps: Azure (Government, ML, OpenAI, AI Document Intelligence, AI Speech, Cosmos DB, Service Bus, Azure SQL, Entra ID), AWS (SageMaker, Lambda, Step Functions, Kinesis), Docker, Kubernetes, MLflow, CI/CD

Experience

Principal Data Scientist, Head of AI Research & Data Science — **Oct 2025 - Present**
Los Angeles County District Attorney's Office — **Los Angeles, CA**

- **Built and lead a first-of-its-kind agentic AI platform for prosecutors**, in production on Azure Government, that **improved case-preparation efficiency by more than 80%** and lets deputy DAs work several complex cases at once: 5,000+ files, multiple defendants, scanned exhibits, transcripts, audio/video, and financial records.
- **Authored TRACE (AAAI-27 manuscript), a harness for workflow-level evaluation of legal agents** that runs each task as a deterministic state machine over typed, source-linked artifacts with claim-level citation checks, tamper-evident audit logs, and staged human approval. In a preregistered fault-injection study, automated verification cut downstream fault propagation from 100% to 18.8% ($p = .008$); the human-gated workflow reached **98.6% citation precision and 6.7% unsupported claims**, versus 84.0% and 20.8% for one-shot RAG.
- **Created claim-level hallucination metrics for grounded legal QA** (Hallucination Risk Score, Legal Evidence Reliance Score) that score each atomic claim for evidence support, citation support, parametric drift, and retrieval weakness. Validated against human labels (AUC 0.91; 250 answers, 1,438 claims), they showed **route-aware RAG cuts hallucination risk by 38% at 16% lower cost,** and reranked hybrid retrieval lifts Hit@k from 0.78 to 0.96.
- **Developed a calibrated hierarchical document classifier** (MPNet with chunk pooling; 11 document types, 6 fraud subtypes) on leakage-controlled grouped splits with selective abstention: **97.2% held-out accuracy, 0.96 macro-F1, 0.028 ECE at 98.6% coverage**, exported to ONNX with exact top-1 parity, so evidence is routed or sent for human review before any model reads it.
- **Ran practitioner research with deputy DAs** to turn manual legal procedures into specifications for seven agentic workflows (charging analysis, statement of facts, transcript summary, case timeline, warrant review), with explicit judgment boundaries, stop conditions, and audit rules, now the platform's 12-stage workflow standard.
- **Designed a filing-recommendation agent for multi-defendant fraud cases** that replaces a 14-step manual process: it builds a case profile across thousands of files, maps facts to statutory elements from a fixed charge catalog, computes limitation dates and restitution, and drafts a court-format memo, pausing for the DA at each legal judgment. Catalog-only charges and a citation on every factual sentence are enforced in code, not left to the model.
- **Built investigative analytics agents:** text-to-SQL over a healthcare-claims database with parser-validated, allowlisted read-only queries that return tables, charts, and financial-flow diagrams; a link-analysis graph (Azure SQL node/edge tables, 1–3 hop traversal) connecting case facts to providers, patients, and payments; and spreadsheet analysis that pairs governed SQL with deterministic statistics and asks clarifying questions mid-query.
- **Engineered ingestion and retrieval at case scale:** layout-aware OCR for scanned TIFF/PDF, transcripts parsed as structured data, audio/video transcription (Azure AI Speech), PII redaction, hybrid retrieval, and task-based model routing across GPT-5.1 and GPT-4o. Context-budget controls turned an 804-file timeline run that failed at 80K tokens into a 52-second run.

- **Secured the agent runtime for government use:** OpenAI Agents SDK behind a framework-neutral orchestration layer, an authenticated MCP server for scoped evidence tools, one-use approval tokens for every write, and hash-based audit events on Azure (App Service, Service Bus, Cosmos DB, Azure SQL, Entra ID, Private Link, Key Vault); Microsoft Graph lets the assistant schedule meetings and send Outlook mail when the attorney approves.
- **Lead AI and data strategy for the office**, working with attorneys, investigators, and IT security, and set the evaluation and governance standards for AI-assisted casework.

Applied Data Scientist **Oct 2024 - Oct 2025**
Microsoft **Remote**

- **Led text analytics on transcripts, documents, and conversations:** fine-tuned LLMs and built RAG pipelines (distributed training on DeepSpeed ZeRO-3, SAFE AI principles) whose insights improved customer interactions.
- **Built Named Entity Recognition and PII-redaction models** in PyTorch, served with ONNX, to process and protect sensitive customer records for compliance.
- **Scaled data processing and model training** on distributed computing frameworks, and worked with Azure AI Language teams on TextAnalyticsClient model development and deployment.

Data Scientist (CTO Office) **May 2024 - Nov 2024, May 2025 - Oct 2025**
Coherent Corp. **Santa Clara, CA**

- **Designed RA-FSM, a research assistant that runs LLMs in a finite-state loop** (relevance, confidence, knowledge) that chooses to retrieve, abstain, or answer, with closed-world citations and claim-to-evidence audits.
- **Evaluated RA-FSM with domain experts against strong baselines:** fewer contradictions and higher reviewer preference at budget-tunable latency and cost; shipped reusable RAG components (retrieval controller, abstention policy, evidence viewer, metrics logger).
- **Built a wafer yield analysis and optimization application** using Failure in Time (FIT) and Defective Parts Per Million (DPPM) metrics, with a time-to-failure regression model (scikit-learn) for power-variation analysis.
- **Increased wafer testing efficiency by 70%** by replacing manual test methods with automated Python tooling.

Machine Learning Engineer **Feb 2024 - July 2024**
Qwarke, Inc. **Remote**

- **Owned the AWS ML pipeline** (Step Functions, Lambda, Kinesis, PySpark, S3, SageMaker) for a scholarly social media platform, with CI/CD for model deployment in an Agile MLOps team.
- **Built a hybrid recommender** (content-based and collaborative filtering) that matched video suggestions to users' fields of expertise, improving engagement.

Projects

CropCatalyst - Reboot the Earth Challenge, United Nations and Salesforce | github.com/vbhavsar16/CropCatalyst

- Predicted crop yield (tons per hectare) from curated agrometeorological data with ensemble models (Random Forest, XGBoost, LightGBM, CatBoost, SVM), and turned yield-gap analysis into native-language advice for farmers.

Flood Semantic Segmentation Model using NVIDIA Toolkit

- Trained a U-Net on satellite radar imagery for flood segmentation in disaster-risk monitoring, with NVIDIA DALI augmentation and TAO on NGC, and deployed it on Triton Inference Server.

Education

University at Buffalo, SUNY Sep 2018 - Jan 2024
Doctor of Philosophy in Computation and Applied Mathematics

University at Buffalo, SUNY Sep 2016 - Aug 2018
Master of Science in Mechanical Engineering

Gujarat Technological University June 2012 - Jul 2016
Bachelor of Engineering in Mechanical Engineering